Ex.no:	DATA VISUALIZATION USING POWER BI (ADVANCED SALES ANALYTICS)
Date:	

Aim:

To Load the data(Retail_sales) into PowerBI and transform the data and visualize it to get insights

Algorithm:

ALGORITHM 1: IMPORTING DATA INTO POWER BI

Step 1

Open Power BI Desktop

Step 2

Select:

Home → Get Data

Step 3

Choose Excel or CSV

Step 4

Browse dataset

Step 5

Click Load

Step 6

Verify imported data

ALGORITHM 2: DATA TRANSFORMATION USING POWER QUERY

Step 1

Select **Transform Data**

Step 2

Open Power Query Editor

Step 3

Remove duplicates

Step 4

Handle null values (replace blanks/nulls in columns like CustomerName and Discount)

Step 5

Rename columns & format text casing (e.g., Capitalize Region)

Step 6

Apply changes & Close

ALGORITHM 3: ADVANCED TRANSFORMATIONS & DATE PARSING

Step 1

Select OrderDate column and ensure data type is set to **Text**

Step 2

Go to **Add Column → Custom Column**

Step 3

Name column: ParsedDate

Step 4

Apply Power Query M Formula:

Code snippet

```
if Text.StartsWith([OrderDate], "2025-") then
    Date.FromText([OrderDate])
else if Text.Contains([OrderDate], "/") then
    Date.FromText([OrderDate], [Format = "dd/MM/yyyy", Culture = "en-GB"])
else
    Date.FromText([OrderDate], [Format = "MM-dd-yyyy", Culture = "en-US"])
```

Step 5

Handle negative quantities (Absolute Value or filter rows)

Step 6

Click **Close & Apply** to load the clean dataset into the model

ALGORITHM 4: CREATING CALCULATED COLUMN (REVENUE)

Step 1

Navigate to **Report View** or **Table View**

Step 2

Right-click dataset table and select **New Column**

Step 3

Enter Revenue formula:

$$Revenue = [Quantity] \times [UnitPrice] \times (1 - (if\ [Discount] = null\ then\ 0\ else\ [Discount]))$$

Step 4

Press **Enter** to validate and compute values

ALGORITHM 5: DASHBOARD VISUALIZATION & REPORTING

Step 1

Select **Line Chart** visualization and map ParsedDate (X-axis) and Revenue (Y-axis)

Step 2

Select **Bar / Column Chart** and map Region (X-axis) and Revenue / Quantity (Y-axis)

Step 3

Select **Donut Chart** to break down quantities by Product

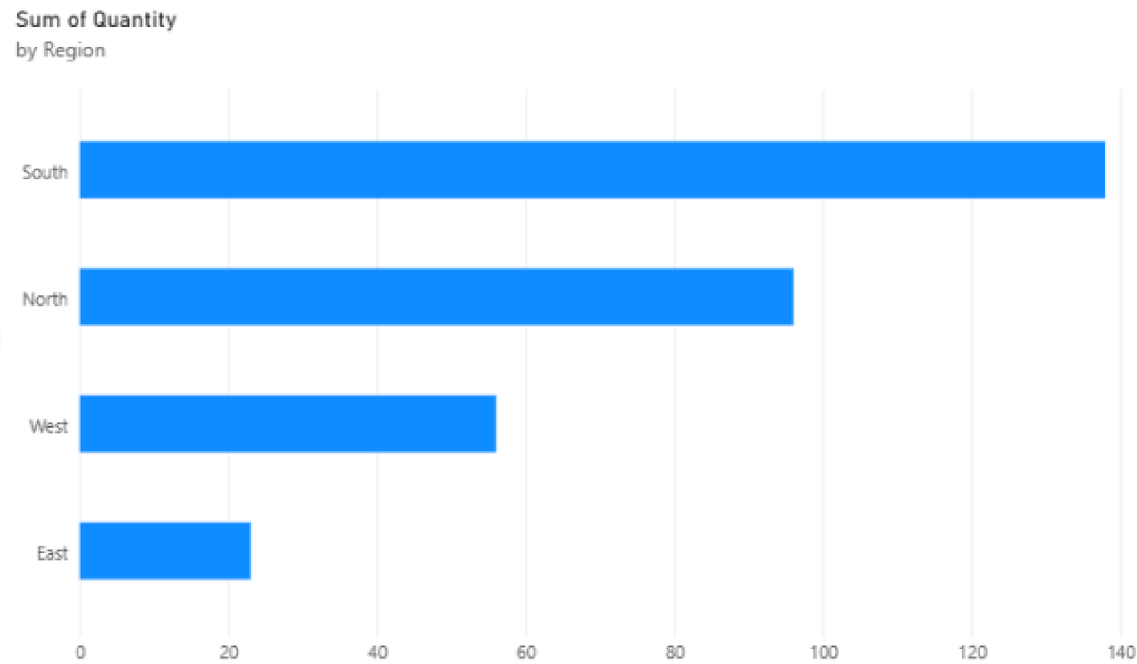
Step 4

Add **Slicer** visuals for Region and Date to enable interactive filtering

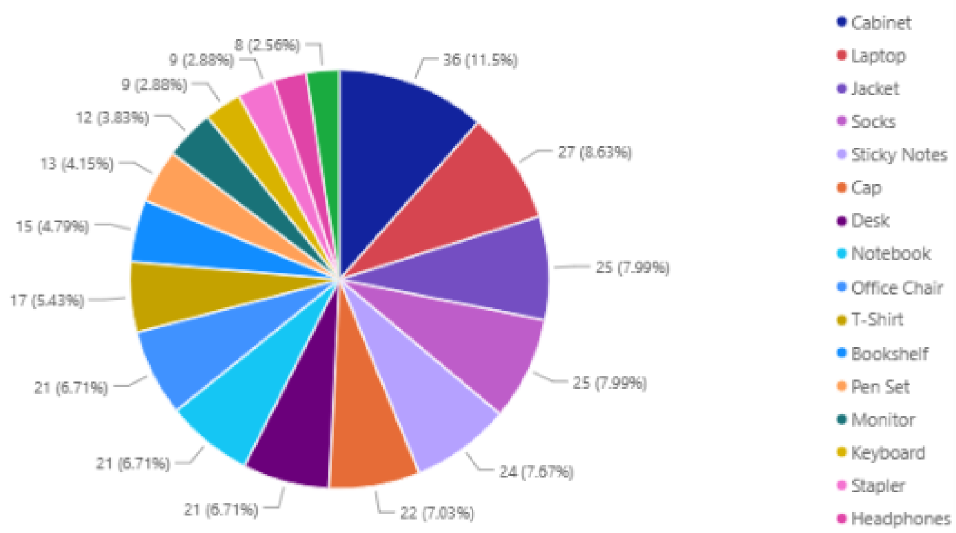
Step 5

Save the Power BI file (.pbix)

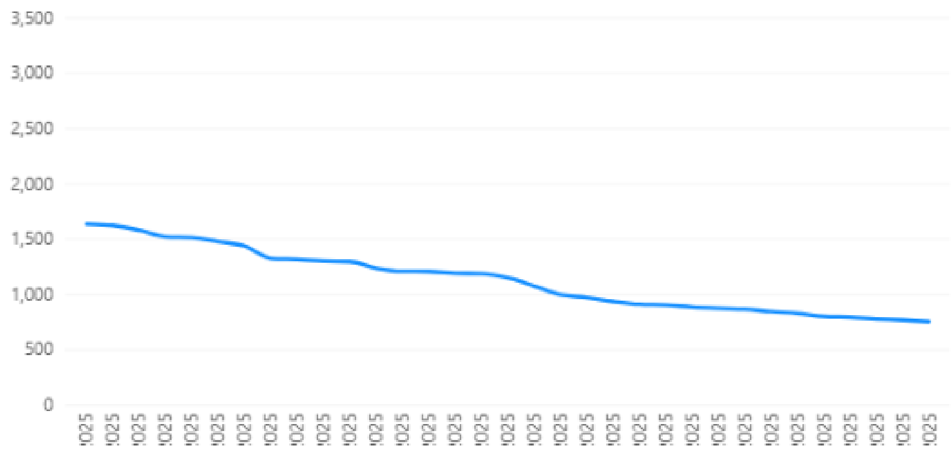
Output:



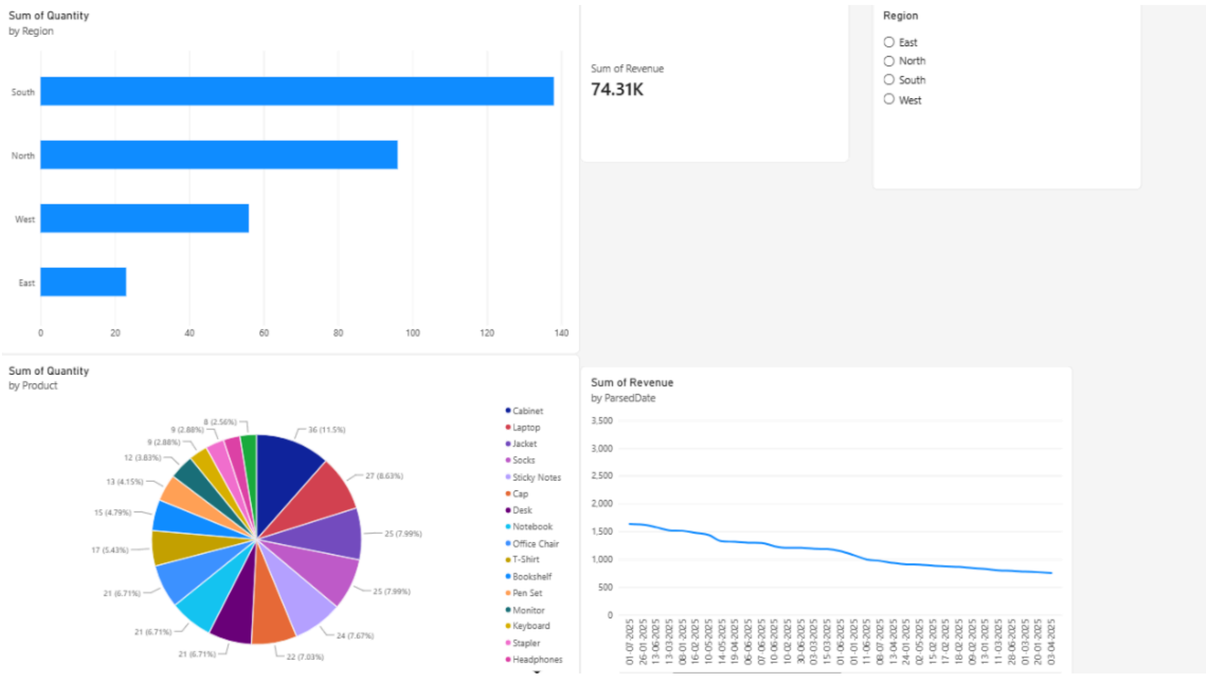
Sum of Quantity
by Product



Sum of Revenue
by ParsedDate




Sum of Revenue
74.31K
Region
East
North
South
West


Sum of Quantity
by Region
South
North
West
East
0
20
40
60
80
100
120
140
Sum of Revenue
74.31K
Region
East
North
South
West
Sum of Quantity
by Product
36 (11.5%)
27 (8.63%)
25 (7.99%)
25 (7.99%)
24 (7.67%)
22 (7.03%)
21 (6.71%)
21 (6.71%)
21 (6.71%)
17 (5.43%)
15 (4.79%)
13 (4.15%)
12 (3.83%)
9 (2.88%)
9 (2.88%)
8 (2.56%)
Cabinet
Laptop
Jacket
Socks
Sticky Notes
Cap
Desk
Notebook
Office Chair
T-Shirt
Bookshelf
Pen Set
Monitor
Keyboard
Stapler
Headphones
Sum of Revenue
by ParsedDate
3,500
3,000
2,500
2,000
1,500
1,000
500
0

Result: